Rock Magnetism and Paleomagnetism of Some Intrusions From Virginia[1]

R. LØVLIE

Department of Geophysics, University of Bergen, Bergen, Norway

N. D. OPDYKE

Lamont-Doherty Geological Observatory, Palisades, New York 10964

A paleomagnetic study of some felsite and basic intrusions in Virginia revealed the presence of presumably stable magnetic directions of normal and reverse polarities in close agreement with the axis of the present geomagnetic field. Results of ore microscopy, Curie temperatures, and X ray diffraction analysis suggest primary titanomagnetites to be the major magnetic mineral present. The paleomagnetic results suggest a Tertiary age of all intrusions in accordance with a radiometrically determined Middle Tertiary age of the felsites.

Concordant felsite outcrops in the Monterey syncline of Virginia are associated with the Devonian Tioga bentonite by a few indications of baked lower contacts but by only one faint indication of a baked upper contact. Field relations alone would thus tentatively suggest the felsite to be flows of Devonian age. A radiometrically determined Eocene age [*Fullager and Bottino*, 1969], however, suggests the felsite to be intrusions. Numerous basic dikes of unknown age, intimately associated with the felsite, may be genetically and temporally related to the felsite.

Paleomagnetism may be used as a dating tool for igneous rocks by comparing the stable magnetic direction in the rock of unknown age with magnetic directions of known age obtained from the same continent. In the present case the method seems especially promising because of the large paleomagnetic differences between the ages concerned.

A total of 12 felsite outcrops were examined during two field trips. Seven outcrops were found to consist of loose blocks that were certainly not in situ and were thus unsuitable for paleomagnetic purposes. Five felsite localities, shown on the sketch map in Figure 1, were sampled by a 2.54-cm motorized diamond drill and by collecting oriented hand samples. A basic intrusion was sampled at two localities.

Megascopically, the felsite samples can be classified as being fresh or weathered; the fresh samples still possess a light gray color, whereas the weathered samples were dark brown (stained by iron hydroxides).

NRM and Susceptibility

The direction and intensity of the magnetic vector were measured by a Par 105-Hz spinner. Low field susceptibilities were measured with an ac bridge.

Histograms of the natural remanent magnetization (NRM) intensities and of bulk susceptibilities for all the 41 felsite specimens are shown in Figure 2. There is a slight tendency for the weathered specimens to have the strongest NRM intensities. The bulk susceptibilities, however, are significantly larger for the weathered specimens, possibly indicating the presence of alteration products having relatively strong specific magnetizations.

Magnetomineralogical Investigations

The opaque minerals were studied by using a Leitz Wetzlar microscope at magnifications up to 1900 diameters. The magnetic colloid technique was used to trace qualitatively the internal magnetic structure within magnetic grains [*Soffel*, 1964]. A total of 11 polished sections from the six localities were investigated.

The results are summarized in Table 1. Generally, the small grain sizes prevented an unambiguous mineral identification. The sites are arranged in descending order according to a qualitative scale of increasing thermal alterations. This scale is defined by increasing amounts of iron hydroxide, possibly goethite (ruby red internal reflections), which apparently starts to replace the magnetic opaques from the center outward. Only specimens from location 6 contained separate ilmenite grains, which appear to be only slightly altered. Some of the ilmenite grains attracted the magnetic colloid, indicating the presence of hemoilmenites with Néel temperatures above room temperature [*Nagata*, 1961]. Figure 3 shows examples of typical titanomagnetite grains belonging to both the least-altered (Figure 3*a*) and the most-altered (Figure 3*b*) states.

The saturation magnetization J_s as a function of temperature T was measured with a translation balance. The heating was done in air in applied magnetic fields ranging between 1500 and 5700 Oe. At least three J_s-T determinations were performed on samples from each locality. It was found that within each locality the J_s-T curves from different samples appeared to show the same features. Curves representing different localities, however, showed dissimilar features. Typical J_s-T curves from the six localities are shown in Figure 4. Some of the Curie temperature data are presented in Table 2. The results are summarized for each site as follows.

Site 1. The J_s-T curves define a single Curie temperature T_c around 550°–575°C that is stable against heating to 700°C, the J_s decreasing by a factor of 0.4–0.6. This is interpreted as being due to high-temperature oxidation of the stable magnetic phase into a magnetic phase with a

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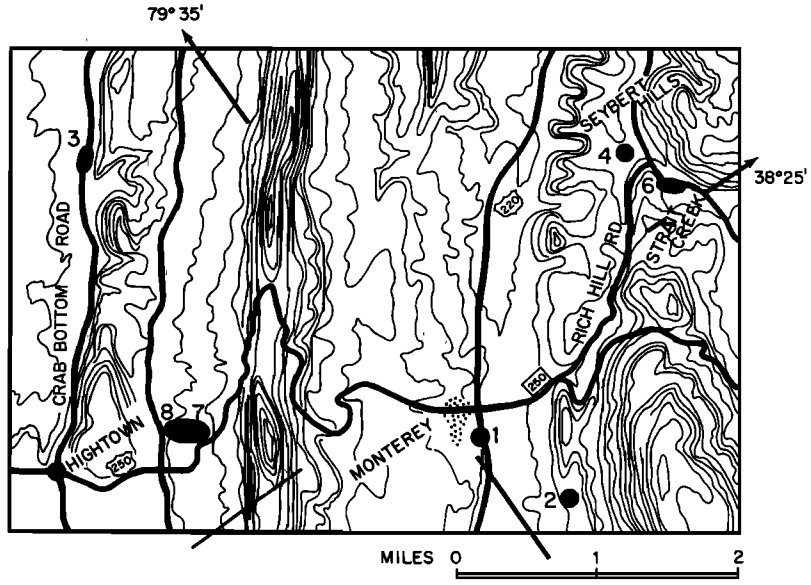


Fig. 1. Sketch map of site localities: 1–6, felsite intrusions; and 7–8, basic intrusions.

lower J_r (Figure 4*a*). The decrease in J_r was found to be a function of both the maximum temperature during the heating and the grain size of the samples. The more finely the samples were crushed, the larger was the decrease in J_r; this finding supports the proposed oxidation mechanism. A T_c around 550°C may represent a magnetite, small amounts of impurities being responsible for the lower Curie temperatures of this range [*Sanver and O'Reilly*, 1970].

Site 2 (Figure 4b). The irreversible J_r-T curves show the typical features for a continuous oxidation/unmixing of a titanomagnetite-maghemite during heating [*Ozima and Ozima*, 1971]. The unstable T_c around 350°C and the increase in J_r after heating support the validity of this interpretation. Ore microscopy confirms the presence of titanomagnetite with maghemite-lined cracks. A typical grain is shown in Figure 3*a*.

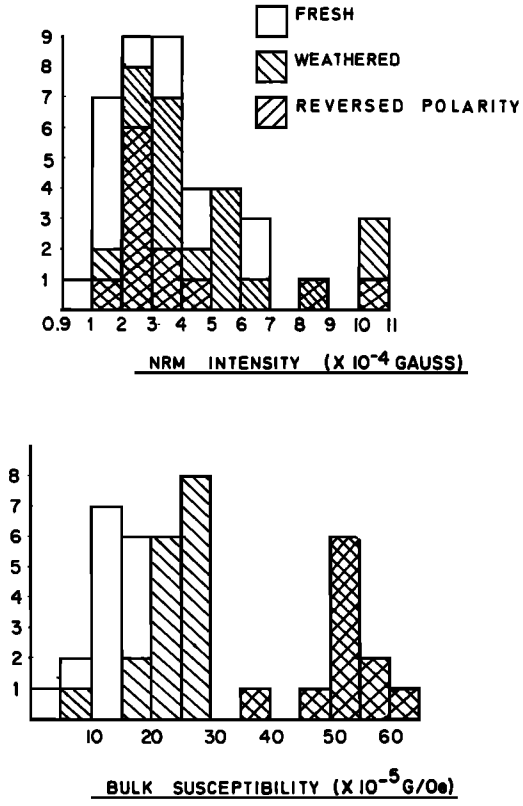


Fig. 2. Histograms of NRM intensities and bulk susceptibilities of 41 felsite specimens.

Site 3 (Figure 4c). The initial T_c around 540°C increases to 575°C after heating to 600°–700°C; this finding suggests the presence of a slightly substituted and/or a cation-deficient form of magnetite that unmixes and forms pure magnetite on heating. The decrease in J_r after heating is shown to be temperature dependent; the dotted line in Figure 4 represents the cooling and heating curve 580°-40°-580°C, whereas the solid line represents the cooling after heating to 700°C. This is interpreted as being due to an oxidation of magnetite into hematite, the amount of the hematite phase being too small for thermomagnetic detection. The 'kink' at around 200°C disappears on cooling after heating to 300°C. Ore microscopy revealed the presence of a large amount of iron hydroxides, possibly goethite. The dehydration of goethite, breaking down to hematite, is reported [*Strangway et al.*, 1967] to start at around 200°C; i.e., the J_r-T data support the results obtained from ore microscopy.

Site 6 (Figure 4d). Two stable Curie temperatures, one around 80°–100°C and one around 540°–560°C, are associated with a slight (10–20%) decrease of J_r after heating. The contribution to J_r from the low Curie temperature phase is unchanged upon heating to 700°C; this finding indicates a magnetic phase with a high thermal stability. As this is certainly not the case for a titanomagnetite with the proper amount of ulvospinel in solid solution [*Ozima and Larson*, 1970], the low Curie point is interpreted to represent a hemoilmenite [*Nagata*, 1961]. This interpretation is in agreement with the results from ore microscopy (Table 1). A substituted magnetite [*Nagata*, 1961] or a stabilized 'maghemite' phase may oxidize to hematite, resulting in a decrease of J_r. As the J_r for hematite is less than 1/200 of the J_r for magnetite [*Nagata*, 1961], this will effectively prevent the detection of the T_c for hematite in this case.

Sites 7 and 8 (Figure 4e). The shape of the J_r-T curves

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TABLE 1. Summary of Ore Microscopical Results of the Magnetic Minerals

Site	Grain Size, μ	Oxidation Class*	Cubic Opaques	Ilm,† %	Colloid§	Sulfide	Alterations‖
7, 8	<10 >20	1	Aneuhedral titanomagnetite.	None	Inhomogeneous.	Traces	None visible.
2	<30	1	Cracked anhedral titanomagnetite. Brighter white at edges.	None	Increasingly magnetic toward edges.	Traces	Maghemitized.
1	5 to 10, <20	1	Aneuhedral cracked titanomagnetite. Bright, white, isotropic.	None	Increasingly magnetic toward edges.	None	Maghemitized iron hydroxides. Hemoilmenite?
6	<30, >100	1	Anhedral cracked titanomagnetite. Isotropic, bright, white. Iron hydroxide relicts of titanomagnetite?	<2	Edges most magnetic.	None	Maghemitized iron hydroxides. Hemoilmenite?
3	5 to 20, <40	1	Aneuhedral cracked titanomagnetite. Hematite coated. Abundant ruby red iron hydroxides replacing cubic phase.	None	Edges most magnetic.	Traces	Maghemitized iron hydroxides. Hemoilmenite?

*High-temperature oxidation class according to *Wilson and Watkins* [1967].
†Hemoilmenite grains.
§Magnetic nature identified by magnetic colloid [*Soffel*, 1964].
‖Low-temperature hydrothermal alterations.

and the low T_c reveal that the magnetic mineral present in the basic rocks is an almost unoxidized form of an ulvospinel-rich (40–50%) titanomagnetite [*Ozima and Larson*, 1970]. The slight increase of J_s upon heating to 700°C indicates a single titanomagnetite phase. Ore microscopy also supports this interpretation; no visible alterations were found on the titanomagnetite grains.

X Ray Diffraction Analysis

X ray diffraction analysis was performed on eight samples. In order to obtain diffractograms with a low noise, the ore minerals were extracted from the powdered samples by using a hand magnet. After grinding this extract in a McCrown micro ballmill to a 10- to 15-μ size, the magnetic extract was enriched by using a technique outlined by *Løvlie et al.* [1972]. Diffractograms were run at 1°–¼°/min, some having Si as an internal standard. Four to six diffraction peaks were used to determine the unit cell size of the cubic phase present in all samples.

Only samples from site 6 showed the presence of a hexagonal mineral phase. The diffraction peaks for the felsite samples showed distinct broadening, often associated with multiple peaks. This broadening is interpreted to be due to the presence of a series of cubic phases with slightly different cell sizes. The sample run from the basic intrusion showed narrow single peaks, confirming the homogeneous single-phase titanomagnetite identified by J_s-T measurements and ore microscopy. The results of both the Curie temperature and the unit cell size determinations are presented in Table 3. The cell sizes range between 8.45 and 8.38 Å, 8.396 Å being indicative of pure magnetite.

Substitution of diamagnetic ions in the magnetite structure may both increase and decrease cell sizes as well as Curie temperatures. The most common diamagnetic ion entering the lattice is titanium, generating the solid-solution series of ulvospinel-magnetite. Increasing amounts of titanium lower the Curie temperature and increase the cell size. Thus the observed larger (than magnetite) cell sizes and proportionally lower Curie temperatures do fit with the titanomagnetite series.

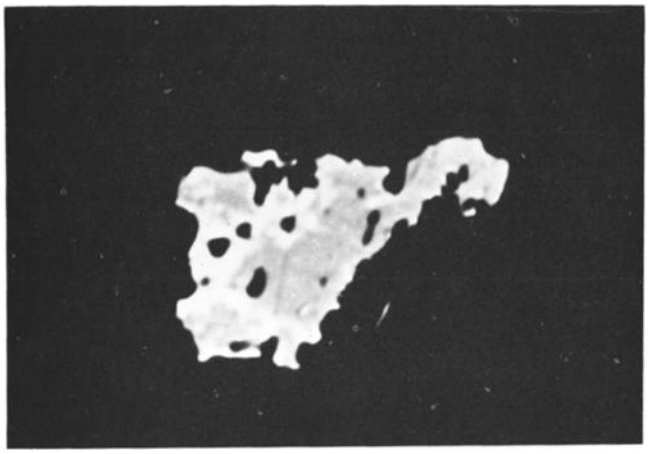
Alternating Field Demagnetization

All drilled specimens and at least two 2.54-cm cylinders from each hand sample were subjected to alternating field (af) demagnetization in steps of 25 and 50 Oe to remove 'soft' magnetic components and to determine the underlying stable directions. A two-axis tumbler rotated in a residual magnetic field of approximately 100 γ, the earth's field being canceled by two pairs of circular Helmholtz coils. It was found that, even with careful field cancelation, all the specimens acquired random magnetic directions, possibly due to an anhysteretic remanent magnetization (ARM) when they were exposed to af fields of 100–250 Oe.

A total of 34 specimens, however, seemed to reach a stable direction after af treatments around 100–150 Oe. The stable directions showed only small directional changes upon successive exposure to increasing alternating fields, associated with a monotonic decrease of the NRM intensity.

The results of the af demagnetization on selected specimens are presented in Figure 5. The stereograms show the stable magnetic directions for individually oriented specimens and typical examples of the directional changes upon successive af demagnetization for a few specimens. The corresponding demagnetization curves are also shown. Specimens from one felsite location (site 4) appeared to have anomalously high Q_n' ratios (25–42), defined as the ratio of the NRM intensity to the magnetization induced by a field of 1 Oe, and divergent stable magnetic directions compared to nearby sites ($Q_n' = 0.5$–5.0). This NRM is interpreted as being caused by lightning strikes and thus is unsuitable for the purpose of the present study. The results

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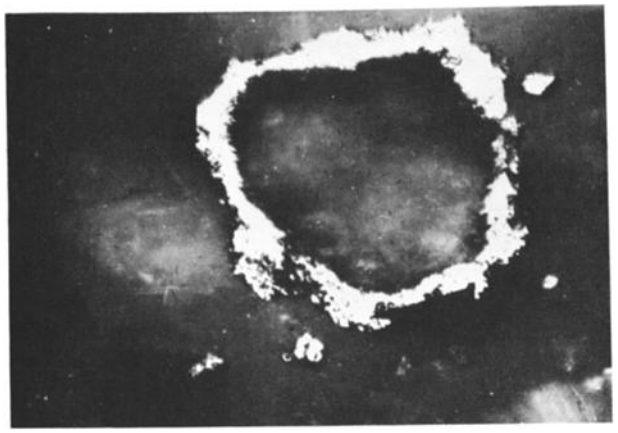


Fig. 3. Typical ore grains. (*a*) Site 2: titanomagnetite (gray) partially altered to titano(?)maghemite (white). (*b*) Site 3: titanomagnetite; center of grain altered to goethite (?).

from the remaining four felsite locations show that two almost antiparallel magnetic polarities are present.

Figure 5*a* shows the stable magnetic directions from two sites (1 and 6) with normal polarity. Samples from site 1 appear to be megascopically fresh compared to samples from site 6, but the af demagnetization curves show similar features. The removal of a large but magnetically soft intensity is associated with a small directional change that reduces the scatter of the NRM directions.

Figure 5*b* shows the stable magnetic directions from sites 2 and 3, both of which show a high degree of alteration (weathering) and which have reversed polarities. The stepwise directional changes for two representative specimens (6a and 7b) upon demagnetization and the intensity decay curves clearly demonstrate that a magnetic field with normal

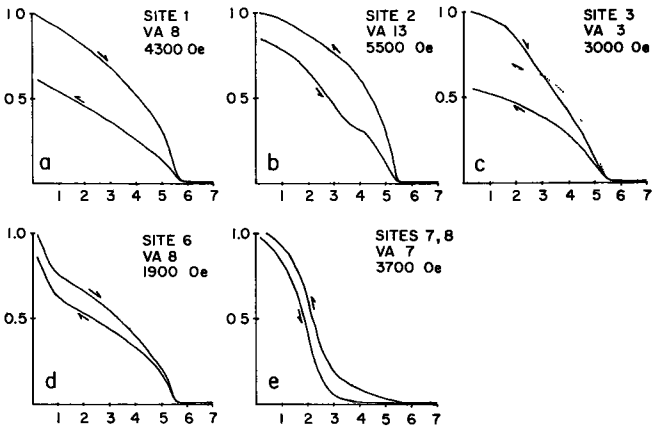


Fig. 4. Normalized J_s-T curves for samples from each site. Numbers on the abscissa represent temperature times 100°C.

polarity has induced large magnetically soft secondary components that seem to be either totally or partly removed in almost all specimens.

The NRM directions for two specimens from site 3 were measured before and after random-oriented storage in the earth's magnetic field for 7 days. Directional differences of 30°–50° were found. Alternating field demagnetization in a 50-Oe field effectively removed the induced component. This finding suggests that the secondary component of normal polarity is a soft viscous remanent magnetization acquired in the earth's present magnetic field.

To test for the presence of a possible self-reversing effect, one specimen from each locality was heated in air to above 680°C and cooled in the earth's magnetic field. The direction of the acquired thermoremanent magnetization (TRM) did not support any self-reversing hypothesis, although a time-temperature sensitive self-reversing mechanism as found in some submarine basalts [*Ozima and Larson*, 1968] cannot be excluded.

The calculated mean declinations, mean inclinations, and relevant statistical parameters [*Irving*, 1964] are given in Table 3.

ORIGIN OF NRM IN TERMS OF MAGNETOMINERALOGY

The initial TRM acquired during the formation of an intrusive rock body will record an average direction of the earth's magnetic field at the site depending on the cooling rate and the Curie points of the magnetic minerals. The magnetic minerals carrying this TRM may later be altered chemically, owing to thermal or hydrothermal activity. Mag-

TABLE 2. Curie Temperatures and Corresponding Unit Cell Sizes of Felsite and Basic Samples

Site	Rock	Sample	Tc_1	Tc_2	R, Å
1	felsite	VA 1	555	555	8.380 ± 0.010
		VA 8	575	575	8.390 ± 0.010
2	felsite	VA 13	350+	565	8.444 ± 0.004
3	felsite	VA 3	535	575	8.406 ± 0.002
		VA 12	535	575	8.400 ± 0.010
6	felsite	VA 5	540/80	540/90	8.388 ± 0.002
		VA 8	545/80	540/90	8.379 ± 0.009
7	basic	VA 7	280	310/530(?)	8.454 ± 0.003

Tc_1 is the initial Curie temperature, and Tc_2 is the temperature after heating to 650° to 700°C; R is the unit cell size of the cubic phase. Plus sign denotes unstable upon heating. Temperature control is ±2°C. Curie temperature definition is ±5°C.

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TABLE 3. Paleomagnetic Results of Felsite and Basic Rocks

Site	Rock	No. of Oriented Samples N	R	Precision Parameter κ	α_{95}	Mean Declination D_{mean}	Mean Inclination I_{mean}	Lat. of Pole Position ϕ, deg	Long. of Pole Position λ, deg
1	felsite	5	4.97	125.90	6.84	356.1	56.3	86.6N	167.7
2	felsite	2	2.00	539.90	10.77	188.4	-48.1	78.6S	240.8
3	felsite	7	6.92	78.35	6.86	176.8	-50.2	82.3S	301.7
6	felsite	11	10.37	15.99	11.78	349.4	67.7	75.5N	253.0
7	basic	6	5.85	32.97	11.84	208.3	-49.3	65.3S	201.2
8	basic	5	4.99	607.52	3.11	171.0	-58.1	82.9S	18.0

R is vector sum of individual directions, and α_{95} is radius of circle of 95% confidence.

netic mineral phases generated by such processes may acquire a thermochemical or chemical remanent magnetization (TCRM or CRM), recording the direction of the ambient magnetic field during their time of formation [*Marshall and Cox*, 1971]. If the primary magnetic minerals are extensively altered, the initial TRM may be replaced completely by secondary magnetizations [*Storetvedt*, 1970].

Ore microscopy investigations show that rocks from different localities have been exposed to different degrees of alteration. Results of X ray and J_s-T curve analysis indicate that primary almost stoichiometric titanomagnetite phases are the major magnetic minerals present.

The stable reversed magnetic directions found in the felsites as well as in the almost unaltered basic intrusions are thus interpreted to be an initial TRM. The statistically significant difference between the radii of the 95% circles of confidence of the two basic rock localities (sites 7 and 8 of Figure 6) from the same igneous body support this conclusion. Site 7 was sampled at the center of the intrusion, where the cooling rate was relatively slow, and thereby a larger part of the geomagnetic secular variation was recorded. This appears as a greater scatter of the individual stable magnetic directions, compared to the much smaller scatter observed from site 8, which was sampled at the border of the intrusion, where the cooling rate was much higher.

The normally polarized samples differ with respect to both the extent of weathering and the magnetomineralogy. The most weathered locality (site 6) contains an ilmenite-hematite phase that does not seem to have been affected by oxidation. The directional changes upon af demagnetization indicate the presence of a reversed component, interpreted to be chemical in origin and carried by the minor amount of secondary formed minerals such as iron hydroxides and maghemite. The slightly oxidized samples from site 1 contain no traces of an ilmenite-hematite phase. In spite of the difference in state of alteration and magnetic mineralogy of the two normally polarized localities, there is good agree-

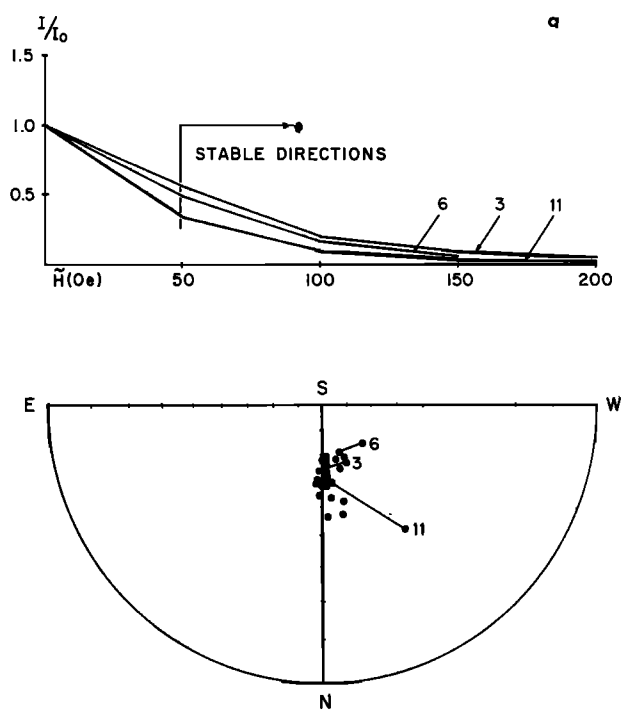


Fig. 5. Directions of stable magnetization and examples of remanence directions and intensities as a function of af fields. (*a*) Felsite, sites 1 (3) and 6 (6, 11).

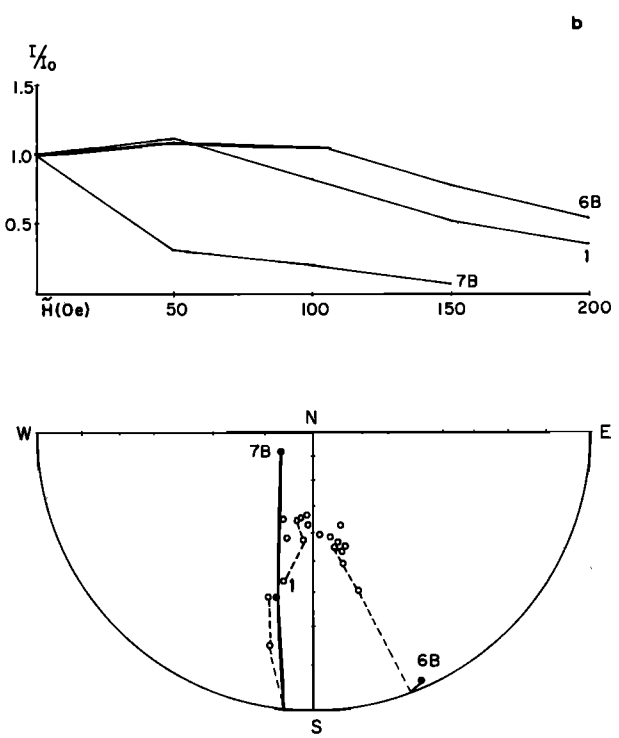


Fig. 5*b*. Felsite, sites 2 (7b) and 3 (1, 6b).

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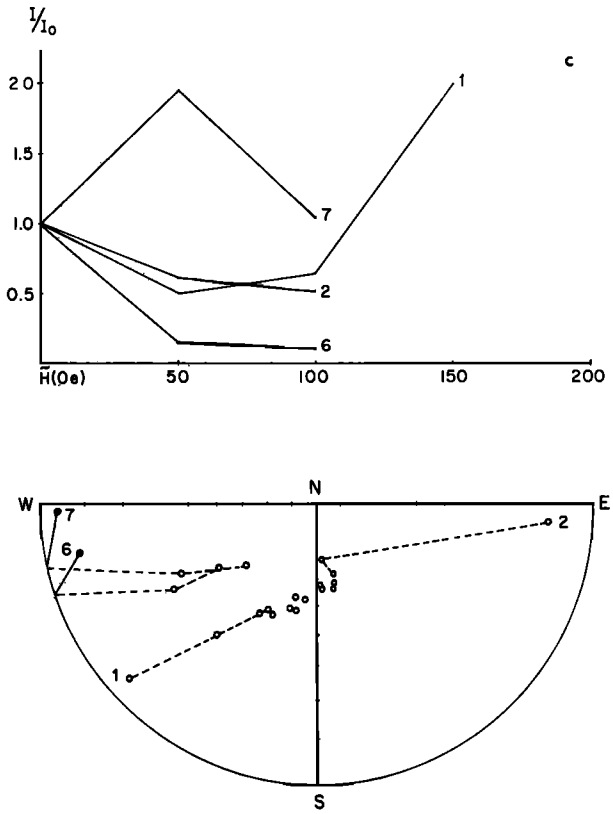


Fig. 5c. Basic intrusion, sites 7 (6, 7) and 8 (1, 2).

ment between the stable magnetic directions. This agreement is interpreted to imply that the stable remanent magnetization is an initial TRM.

Discussion and Conclusion

The stable paleomagnetic directions obtained define an axis that coincides fairly closely with the axis of the earth's present magnetic field. The presence of almost stoichiometric titanomagnetites implies that the stable remanent magnetization is primary, thus indicating a Tertiary-Recent magnetic age for both the felsites and the basic intrusions. A Devonian age for the intrusions is not possible for two reasons.

1. The directions of magnetization are grossly different from any known stably magnetized Paleozoic rocks from North America [*Irving*, 1964].

2. The sites are situated in rocks folded during the Permian orogeny. Restoration to the paleohorizontal serves to scatter the directions of magnetizations. The rocks clearly must postdate the Appalachian orogeny. A pole position calculated from these rocks would not be significantly different from pole positions from early Tertiary igneous rocks of the western United States [*Larson et al.*, 1969]. The results are in accord with the Eocene age previously determined by *Fullager and Bottino* [1969], confirming the intrusive origin of the felsites.

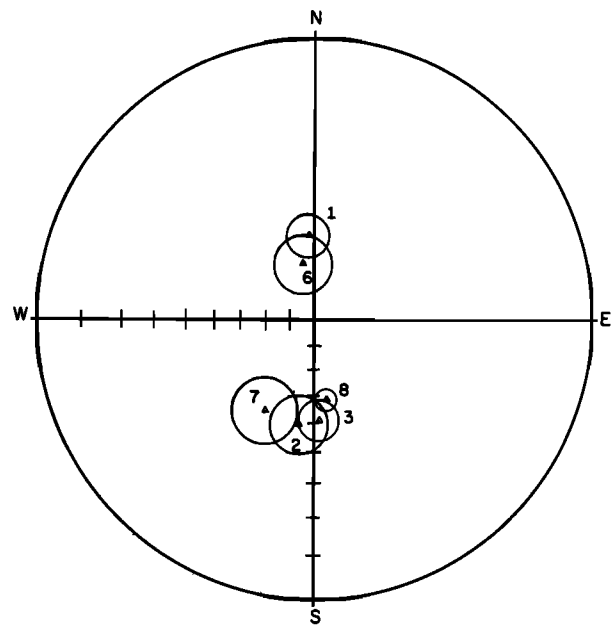


Fig. 6. Site mean direction with 95% circles of confidence. Numbers represent the sites.

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